

LEXOCRATES

Software Requirements Specification

AI-native Legal Operations ERP Platform

BASELINE Client Registration → Client & Portal Users → Wallet / Quoted Billing → Matters → Jobs → Executable Workflows & SOPs → AI + Human Review → QA / Compliance → Delivery

Document ID	Version	Status	Date
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Owner	Audience	Classification
Lexocrates Product & Engineering	Engineering, QA, Security, Operations, Product	Confidential

Prepared as an implementation baseline. Business, legal, tax and regulatory controls marked “policy-dependent” require validation by the responsible Lexocrates owner before production release.

Document Control

Version	Date	Author / Owner	Change summary	Approval
0.1	2026-08-11	Product Architecture	Initial consolidated draft	Pending
1.0	2026-08-11	Lexocrates	Development baseline: complete platform scope	To be approved

Approval Record

Role	Name	Decision	Date	Comments
Product Owner		Pending		
Engineering Lead		Pending		
Security / Privacy		Pending		
Operations Owner		Pending		
Finance Owner		Pending		

Change Control Rules

- Requirement IDs are permanent after baseline approval; retired requirements remain listed with status “Deprecated”.
- Every change request records rationale, affected requirements, data migration, security impact, test impact, approver and target release.
- Workflow, SOP, prompt and policy definitions use independent semantic versions and immutable published snapshots.
- Production deployments must reference a release tag and migration identifier; rollback instructions are mandatory.

Conventions

Term	Meaning
SHALL	Mandatory, testable requirement.
SHOULD	Recommended requirement; deviation requires documented rationale.
MAY	Optional capability.
P0 / P1 / P2	Must-have / should-have / later enhancement priority.
Tenant	A Client organization and its logically isolated data boundary.

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Page references correspond to this Version 1.0 baseline. In the editable Word source, update this table when sections are added or repaginated.

1. Introduction

1.1 Purpose

This SRS defines the functional, data, interface, security, audit and quality requirements for the Lexocrates ERP platform. It begins when an organization registers as a prospective client and continues through identity provisioning, commercial setup, matter and job execution, AI orchestration, human legal production, quality assurance, delivery, billing and long-term audit retention.

1.2 Product Vision

Lexocrates is an AI-native, SOP-driven legal production platform. ERPNext/Frappe provides the authoritative transactional core; a branded client portal hides ERP terminology; a configurable Legal Workflow Orchestration Studio expresses executable business processes; n8n handles external automation; Mattermost supplies matter-linked collaboration; and an AI gateway provides governed model access.

DECISIONS FIXED BY THIS BASELINE The organization is the Client; a Client may have [REDACTED] many Portal Users. Contacts do not automatically receive login access. “Engagement” is replaced by “Matter”. Matters are billed by Quoted Price or Wallet (LexPack). LexPoints never expire. No “trial matter” entity exists.

1.3 Scope

- Public registration and assisted registration by staff
- Client onboarding, KYC/conflict review and activation
- Multi-user client portal with organization- and matter-level permissions
- Quoted-price and non-expiring LexPoint wallet billing
- Matter, Job, Task, Document, Deliverable and communication lifecycles
- Versioned workflows, SOP execution, prompts and AI orchestration
- Role-specific dashboards and legal workspaces
- ERPNext implementation, integrations, observability, security, audit and continuity

1.4 Out of Scope for Baseline

- Autonomous legal advice without required human approval
- Trust accounting unless separately approved by finance/legal counsel
- Country-specific tax determination beyond configured ERPNext tax rules
- A native office-suite replacement; document co-editing may be integrated
- Training foundation models on client data without explicit contractual and privacy authorization

1.5 Stakeholders

Stakeholder	Primary interest
Client organization	Confidential, predictable, transparent legal service
Portal users	Simple matter-centric experience and scoped access
Operations	Throughput, SLA, capacity and intervention
Legal production	Guided work, research, review and version control
QA / Compliance	Evidence, risk gates, calibration and auditability
Finance / Sales	Quotes, wallet, invoicing and profitability
Platform administration	Secure, reliable and governable infrastructure
Engineering / QA	Unambiguous, testable, maintainable baseline

2. System Context and Architecture

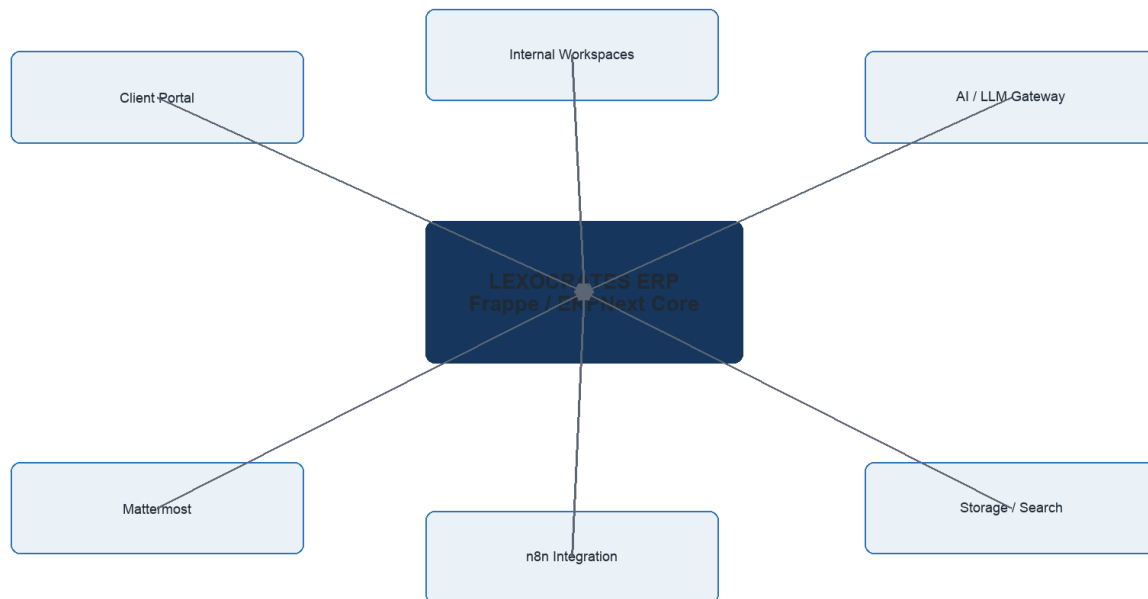


Figure 1. Logical system context and integration boundary

2.1 Architectural Principles

- Single source of truth: ERPNext/Frappe is authoritative for clients, matters, jobs, wallet transactions, approvals and audit references.
- Tenant isolation: every tenant-owned record carries an immutable Client identifier; authorization is enforced server-side.
- Configuration over customization: workflows, SOPs, routing rules, templates and notifications are versioned configuration.
- Immutable financial ledger: wallet balance is derived from posted transactions; posted entries are reversed, never edited.
- Human accountability: AI may propose, extract, classify and draft, but approval gates are dictated by risk and policy.
- Asynchronous by design: AI, conversion, OCR, indexing and external integrations use idempotent jobs and observable queues.
- Zero-trust interfaces: external and internal calls are authenticated, authorized, rate-limited and correlated.

2.2 Logical Components

Component	Responsibility	System of record
Client Portal	Branded self-service UI, submissions, approvals and reports	ERPNext APIs
ERPNext / Frappe	Transactional domain, roles, workflows, ledger, reporting	Yes
Workflow Runtime	Versioned graph execution, timers, branches, gates	ERPNext + execution store
SOP Engine	Checklist instances, evidence, gating and attestations	ERPNext
AI Gateway	Model routing, policy, prompts, cost, safety and telemetry	AI execution records
n8n	External adapters and non-core automation	No
Mattermost	Matter-linked collaboration and notifications	Messages in Mattermost; references in ERP
Object Storage / Search	Encrypted files, versions, OCR text and index	Document metadata in ERP

2.3 Deployment View

Recommended production topology separates web/API, background workers, database, Redis/queues, object storage, search/vector services, AI gateway, n8n and Mattermost. Production and non-production environments must use separate credentials, data stores and encryption keys. Private services should not be internet-addressable.

3. Domain Model and Terminology

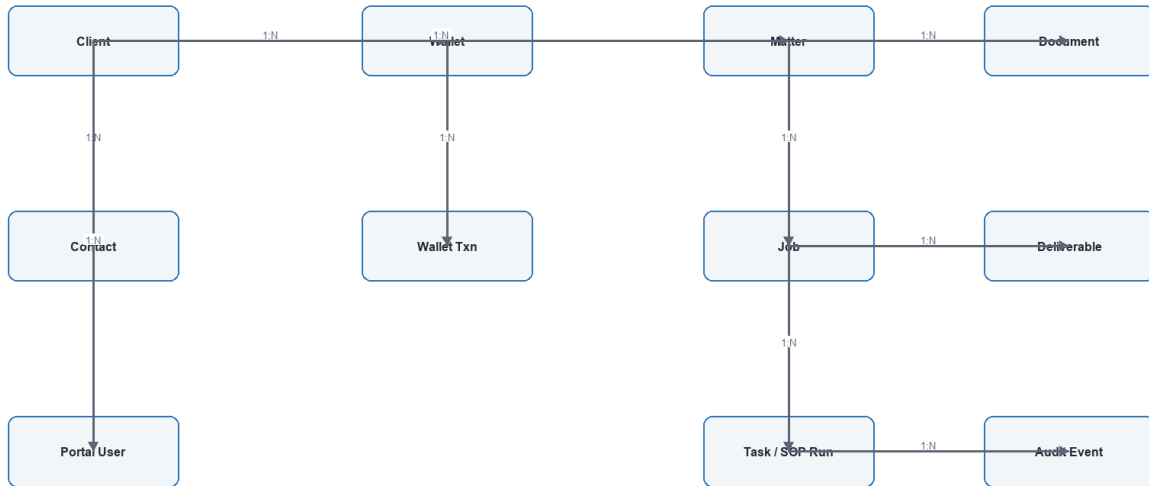


Figure 2. Core entity relationships (simplified)

3.1 Canonical Definitions

Term	Definition
Client	A legal organization purchasing or evaluating Lexocrates services; the tenant root.
Person	A person related to a Client who may use Lexocrates services but has no login by default.
Portal User	An authenticated identity linked to a Client and normally to a Contact; may hold multiple roles.
Matter	A specific legal assignment, project or case. Replaces the ambiguous term Engagement.
Job	An executable work unit within a Matter, independently assigned and progressed.
Task	An atomic human or system step created by a workflow execution.
SOP	A versioned standard operating procedure containing instructions, checklist items, evidence rules and gates.
LexPack	The commercial product that adds LexPoints to a Client Wallet.
Wallet	The internal accounting object that stores non-expiring LexPoint balance via a transaction ledger.
Deliverable	A versioned, approved output released to the Client.
Workflow Definition	A versioned executable graph; published versions are immutable.
Workflow Execution	A run pinned to one workflow version and associated with a Matter or Job.

3.2 Cardinality and Ownership Rules

- One Client has zero or many [REDACTED], Portal Users, Departments, Matters and Wallet Transactions.
- A Portal User belongs to exactly one Client in baseline; cross-client access requires an explicit internal role or future federation model.
- [REDACTED]. A Portal User must reference a verified email identity and a Client.
- A Matter belongs to one Client and has one billing method at a time. Changes after financial posting require an approved adjustment workflow.
- A Matter has one or more Jobs; a Job may have many Tasks, documents, AI runs, reviews and deliverables.
- Every tenant-owned object must inherit its Client ID directly or through a server-validated parent relationship.

4. Roles, Permissions and Data Segregation

Role	Scope
System Administrator	Platform configuration, identity, secrets, infrastructure and global audit.
CEO / Executive	Enterprise KPIs, finance and risk; no routine system administration.
Operations Manager	Broad production visibility and intervention; no secrets or platform configuration.
Sales / Client Success	Leads, quotes, onboarding, wallet sales and relationship management.
Finance	Invoices, payments, wallet posting, reconciliation and reporting.
Senior Associate	Legal review, AI validation, junior approval and knowledge contribution.
Junior Associate	Guided SOP-based production within assigned Jobs.
QA Manager / Reviewer	Quality review, calibration, error taxonomy and release gate.
Compliance / Privacy	KYC, conflict, privacy, retention, incidents and regulatory evidence.
Client Administrator	Manage own organization users, departments and allowed settings.
Client Legal User	Create/view authorized matters, documents, approvals and messages.
Client Finance User	Invoices, payments, quotes and wallet; no confidential documents by default.
Client Auditor	Read-only access to explicitly shared matters and reports.

4.1 Permission Matrix

Capability	Sys Admin	Ops	Legal	QA	Finance	Client Admin	Client User
Manage platform config	A	R	-	-	-	-	-
View all production	R	A	Scoped	Scoped	Financial	Own tenant	Scoped
Create matters	Support	A	Assist	-	-	Yes	Yes
Assign jobs	Support	A	Lead	-	-	-	-
Approve legal output	-	Override*	A/R	A/R	-	Client gate	Client gate
Post wallet transaction	-	Request	-	-	A/R	Purchase	View
Manage client users	Support	-	-	-	-	A/R	-
View API secrets	A/R	-	-	-	-	-	-
View audit	A	Operational	Own work	Quality	Financial	Own tenant	Own actions

Legend: A = accountable/configure; R = execute; Scoped = assigned or explicitly granted; “-” = no access.

*Operations override must be separately permissioned and audited; it cannot substitute for legal or compliance approval when policy requires those gates.

4.2 Authorization Rules

ID	Requirement	Priority	Verification / acceptance
SEC-001	The system SHALL deny access by default and evaluate role, Client scope, matter grant, data classification and action before returning data.	P0	Cross-tenant and unauthorized-object tests return 403 without data leakage.
SEC-002	The system SHALL enforce permissions in server-side APIs and background jobs, not only in the user interface.	P0	Direct API calls cannot bypass UI restrictions.
SEC-003	Client Administrators SHALL manage users only inside their Client and SHALL NOT grant permissions they do not possess.	P0	Privilege-escalation test suite passes.
SEC-004	Matter-level grants SHALL support viewer, contributor, approver and finance scopes with effective-from/to dates.	P0	Access changes take effect immediately and are audited.
SEC-005	Internal support access to a Client SHALL be time-bound, reason-coded and auditable.	P0	Expired access is rejected; audit contains requester, approver and reason.
SEC-006	Sensitive finance, health, identity and privileged content SHALL support additional field/file restrictions.	P1	Configured classifications alter access and download behavior.

5. Client Registration and Onboarding

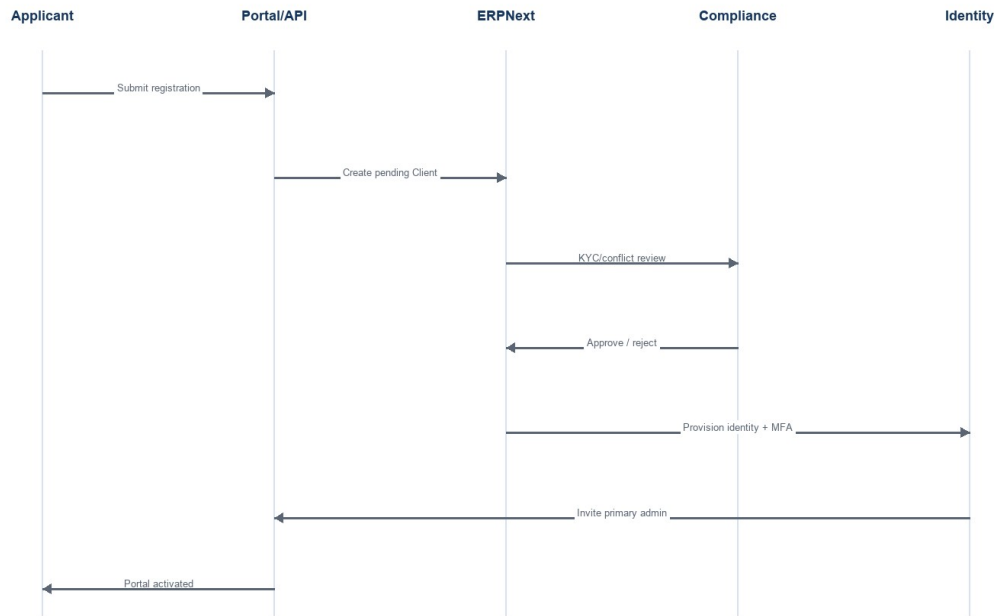


Figure 3. Client registration and activation sequence

5.1 Registration Data

Group	Fields / rules
Organization	Legal name, trading name, entity type, country, registration number, website, industry
Primary contact	Name, title, business email, phone, preferred language and timezone
Legal / compliance	Jurisdictions, conflict parties, service interests, NDA request, privacy consent
Commercial	Preferred currency, billing address, tax IDs, purchase-order requirements
Security	Domain, SSO interest, data residency need, MFA enrollment on activation
Evidence	Consent timestamps, terms version, privacy notice version, source, referrer and IP risk metadata

5.2 Registration States

State	Meaning	Allowed transition
Draft	Applicant has not submitted	Submitted / Abandoned
Submitted	Immutable intake snapshot created	Under Review / Rejected
Under Review	KYC, conflict and commercial checks active	Needs Information / Approved / Rejected

State	Meaning	Allowed transition
Needs Information	Applicant response required	Under Review / Withdrawn
Approved	Controls satisfied; provisioning authorized	Provisioning
Provisioning	Client, identity, folders and settings being created	Active / Provisioning Failed
Active	Portal enabled and work may be submitted	Suspended / Closed
Suspended	Login or transactions restricted	Active / Closed
Rejected / Closed	Terminal business state	Reopen only by authorized workflow

5.3 Functional Requirements

ID	Requirement	Priority	Verification / acceptance
REG-001	The public registration form SHALL validate required fields, email ownership, consent and duplicate organization indicators before submission.	P0	Invalid or duplicate submissions produce actionable messages and no duplicate active Client.
REG-002	Submission SHALL create a pending Client record and immutable intake snapshot, not a Matter, Wallet transaction or active user.	P0	Database inspection confirms only permitted pending objects exist.
REG-003	The system SHALL route registrations through configurable KYC, conflict, sanctions and commercial-review tasks.	P0	Review tasks and owners are created from the active onboarding workflow version.
REG-004	Reviewers SHALL request information without exposing internal notes, conflict logic or other clients.	P0	Applicant sees only the sanitized request and can respond securely.
REG-005	Approval SHALL provision Client defaults, primary Client Administrator invitation, billing profile, security profile, document root and optional empty Wallet.	P0	Provisioning is idempotent and produces one of each configured resource.
REG-006	The system SHALL separate Contacts from Portal Users; enabling login requires an explicit invitation and verified identity.	P0	Contact creation alone cannot authenticate.
REG-007	Invitations SHALL expire, be single-use and require MFA enrollment according to policy.	P0	Reused/expired tokens fail; successful acceptance records MFA state.
REG-008	Client Administrators SHALL invite, suspend and revoke users and assign only approved client-side role templates.	P0	Tenant-scoped administration tests pass.
REG-009	Rejected or abandoned registrations SHALL follow configurable retention and deletion rules.	P1	Scheduled retention job produces evidence and deletion/anonymization log.
REG-010	Registration and onboarding SHALL be available through assisted internal entry using the same validation and audit controls.	P1	Staff-created registration follows equivalent review steps.

6. Client, Contacts and Portal Users

6.1 Client Profile

The Client is the tenant root and represents an organization, not an individual. The profile includes identity, addresses, jurisdictions, departments, billing and tax settings, service preferences, client playbook, AI restrictions, data residency, retention and notification settings.

6.2 Contact and User Lifecycle

ID	Requirement	Priority	Verification / acceptance
USR-001	The system SHALL allow multiple Contacts and multiple Portal Users per Client.	P0	At least 100 users can be linked to one Client without changing Client identity.
USR-002	A Portal User SHALL reference a Client and SHOULD reference a Contact to avoid duplicate person data.	P0	Integrity rule rejects a portal user with no Client.
USR-003	Email uniqueness SHALL be enforced according to the selected identity-provider model; any exception SHALL use verified federation identifiers.	P0	Duplicate login identities are blocked or intentionally federated.
USR-004	User status SHALL include Invited, Active, Suspended, Locked and Revoked.	P0	Only Active users authenticate; status changes are audited.
USR-005	Department and cost-center attributes SHALL be available for routing, reporting and matter access.	P1	Rules can reference department/cost center.
USR-006	Client Administrators SHALL delegate administration for a defined period and scope.	P1	Delegated rights expire automatically.
USR-007	The platform SHALL show active sessions and allow authorized revocation.	P1	Revoked session cannot refresh tokens.
USR-008	User deletion SHALL preserve authored audit events while removing or anonymizing personal data when legally permitted.	P0	Deletion report distinguishes retained evidence and erased profile fields.

7. Commercial Model: Quotes, LexPacks and Wallet

7.1 Principles

COMMERCIAL RULE A Matter is always a Matter. Its Billing Method is either Quoted Price or Wallet (LexPack). A Client may alternate freely between methods. LexPoints never expire and are not hours.

7.2 Wallet Ledger

Transaction type	Sign	Examples	Posting behavior
Purchase / Top-up	Credit	LexPack sale, promotional grant	Posted after approved payment/credit policy
Reservation	Memo / hold	Expected job consumption	Reduces available, not ledger balance
Consumption	Debit	Completed milestone or approved usage	References Matter, Job and rate version
Release	Memo / hold reversal	Cancelled or lower actual use	Restores available amount
Adjustment	Credit or debit	Authorized correction	Separate approved entry; never edit original
Refund	Debit	Commercial refund	Requires finance approval and payment reference
Reversal	Opposite sign	Void erroneous posting	Links original and reason

7.3 Functional Requirements

ID	Requirement	Priority	Verification / acceptance
WAL-001	The system SHALL maintain at most one active Wallet per Client per LexPoint unit scheme.	P0	Duplicate active wallet creation is rejected.
WAL-002	LexPoints SHALL have no expiry date; the schema and UI SHALL NOT require or imply expiry.	P0	Balance remains unchanged across time-based jobs; no expiry scheduler exists.
WAL-003	Posted Wallet Transactions SHALL be immutable and corrected only by linked adjustment or reversal entries.	P0	Update/delete of posted entries is blocked.
WAL-004	Displayed balance SHALL reconcile to the sum of posted credits and debits; available balance SHALL also subtract active reservations.	P0	Automated reconciliation matches ledger for every Wallet.
WAL-005	Each consumption SHALL reference Client, Wallet, Matter, Job where applicable, amount, unit, rate card version, actor, timestamp and idempotency key.	P0	Incomplete posting fails validation; duplicate key does not double-debit.
WAL-006	Wallet-funded work SHALL not start when available balance is	P0	Boundary tests enforce balance or

ID	Requirement	Priority	Verification / acceptance
	insufficient unless an authorized overdraft policy applies.		approved exception.
WAL-007	The portal SHALL show opening balance, credits, debits, reservations, available balance, descriptions and downloadable statements.	P0	Statement totals reconcile and tenant filters hold.
WAL-008	Top-up purchases SHALL support quotation/order/payment integration and post credit exactly once.	P0	Repeated payment webhook results in one credit.
WAL-009	Quoted-price Matters SHALL support quote versioning, client acceptance, scope, tax, currency and change orders.	P0	Execution gate respects accepted quote and approved changes.
WAL-010	Switching a Matter billing method after posting SHALL require finance approval and create traceable adjustments.	P1	No silent reassignment of historical transactions.
WAL-011	Wallet alerts SHALL be configurable by absolute and forecast thresholds and SHALL never present a false expiry warning.	P1	Threshold notifications fire once per policy window.
WAL-012	LexPoints SHALL be described as service capacity units, not time, in portal-facing text.	P0	Content audit finds no hourly-equivalence claim unless contractually configured.

8. Matter Management

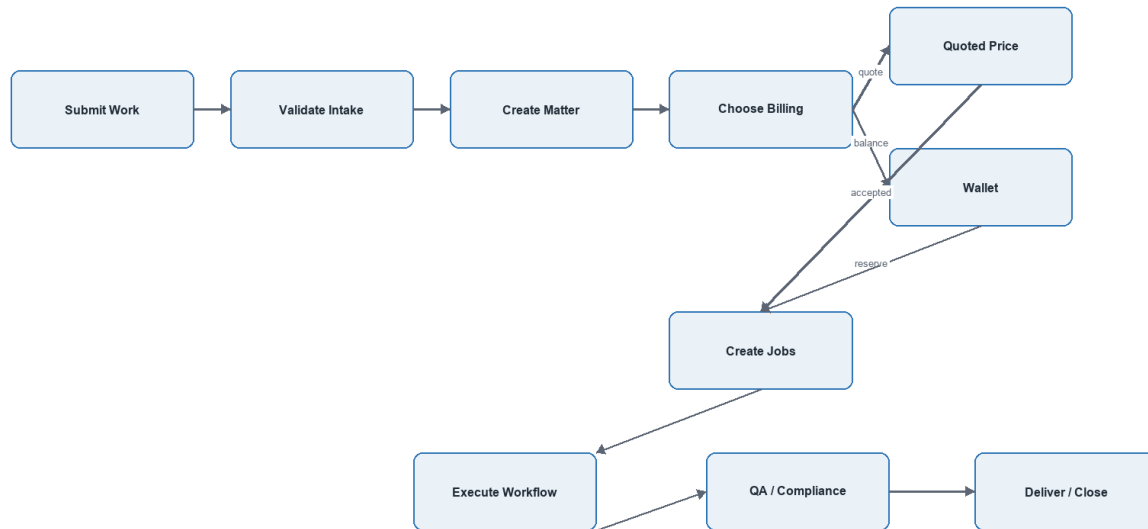


Figure 4. Matter intake, billing and production process

8.1 Matter Attributes

Category	Required attributes
Identity	Matter ID, Client, title, description, external reference
Classification	Practice area, work type, jurisdiction, confidentiality, privilege, risk
Ownership	Relationship manager, operations manager, responsible lawyer, client contacts
Service	Priority, requested/committed dates, SLA policy, workflow template
Commercial	Billing method, quote/rate version, wallet/reservation, PO/cost center
Governance	Status, hold reason, conflicts outcome, retention, legal hold, access groups

8.2 Matter States

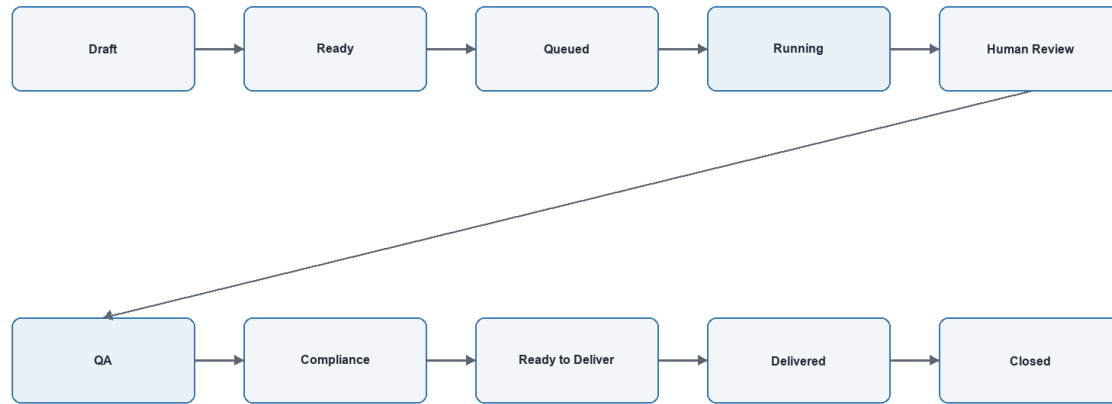
State	Entry condition	Exit / controls
Draft	Intake saved	Submit or discard
Intake Review	Required fields/files present	Clarification, quote or approval
Awaiting Quote	Quoted-price proposal issued	Accept, revise, reject

State	Entry condition	Exit / controls
Acceptance		
Awaiting Funding	Wallet balance/reservation insufficient	Top-up or authorized exception
Open	Commercial and conflict gates passed	Create/execute Jobs
On Hold	Client, legal, compliance or operational hold	Authorized resume
Delivery Pending	Jobs complete; release approval pending	Deliver or return
Completed	Delivery accepted and financial postings complete	Close
Closed	Retention begins; read-only by default	Reopen via approval
Cancelled	Authorized termination	Financial reservations released

8.3 Functional Requirements

ID	Requirement	Priority	Verification / acceptance
MAT-001	The system SHALL create a unique, non-reused Matter ID and link the Matter to exactly one Client.	P0	Concurrent creation produces unique identifiers.
MAT-002	Matter submission SHALL validate mandatory metadata, files, billing method, conflicts status and requested delivery.	P0	Incomplete submission remains Draft with field-level errors.
MAT-003	Quoted-price Matters SHALL require an accepted quote or authorized exception before production begins.	P0	Workflow start is blocked without gate evidence.
MAT-004	Wallet Matters SHALL require a sufficient reservation or approved exception before chargeable execution.	P0	Available balance and reservation update atomically.
MAT-005	A Matter SHALL support multiple Portal Users with viewer, contributor, approver and finance permissions.	P0	Matter grants restrict lists, search, files and APIs.
MAT-006	Matter timeline SHALL record material client, system, AI, human, billing and delivery events in chronological order.	P0	Timeline matches underlying records and filters sensitive internal events.
MAT-007	The system SHALL support configurable client instructions and playbooks inherited by Jobs with explicit override precedence.	P1	Resolved instruction snapshot is stored on Job execution.
MAT-008	Matter holds SHALL stop configured timers, task dispatch and delivery while preserving evidence.	P0	Hold/resume behavior follows SLA policy and is audited.
MAT-009	Matter closure SHALL verify completed/cancelled Jobs, released reservations, final delivery, invoice state and retention classification.	P0	Closure checklist blocks unresolved mandatory items.
MAT-010	Matter search SHALL support permitted metadata, full-text documents and timeline events without cross-tenant leakage.	P0	Search authorization tests pass.

9. Job, Task, Document and Deliverable Management



Exception paths: Failed -> Retry / Manual Triage | Rejected -> Prior Review | Cancelled -> Terminal

Figure 5. Job lifecycle and exception paths

9.1 Job Decomposition

A Matter may be decomposed manually, from a template, or by AI-assisted classification into independently executable Jobs. Examples include NDA review, research memorandum, due-diligence batch, chronology, translation, citation verification and final report assembly.

9.2 Functional Requirements

ID	Requirement	Priority	Verification / acceptance
JOB-001	Each Job SHALL have a unique ID, Matter, Client, work type, owner, priority, SLA, workflow version and status.	P0	Creation fails without required associations.
JOB-002	Job creation SHALL support templates, bulk generation and reviewed AI recommendations.	P1	Generated jobs remain reviewable before dispatch when policy requires.
JOB-003	Assignment SHALL consider skills, jurisdiction, capacity, conflicts, availability and separation-of-duties rules.	P1	Ineligible assignees are excluded or require authorized override.
JOB-004	Every state transition SHALL validate permissions, predecessor tasks, mandatory SOP items and evidence.	P0	Invalid transition returns reason and remains unchanged.
JOB-005	Jobs SHALL support pause, resume, reassignment, escalation, cancellation, retry and return-for-rework.	P0	Each action updates timers and creates audit/timeline events.
JOB-006	The system SHALL maintain optimistic locking or equivalent protection against conflicting edits.	P0	Concurrent edit test detects stale update.

ID	Requirement	Priority	Verification / acceptance
JOB-007	SLA clocks SHALL use configured calendars, timezones, pauses and escalation thresholds.	P0	Known calendar examples compute correct due times.
JOB-008	Documents SHALL support immutable versions, checksums, source, classification, virus scan, OCR status and access scope.	P0	Download retrieves the selected verified version; tampering is detected.
JOB-009	Derived files SHALL preserve lineage to source versions and processing executions.	P0	Lineage graph identifies source, tool/model and timestamp.
JOB-010	Deliverables SHALL require the configured legal, QA, compliance and client approvals before release.	P0	Release API rejects missing gates.
JOB-011	Delivery SHALL record recipient, channel, exact file versions, checksum, timestamp and acknowledgement status.	P0	Delivery receipt can be reproduced from records.
JOB-012	Rejected deliverables SHALL create a reason-coded rework cycle without overwriting prior versions.	P0	Prior delivered/reviewed versions remain accessible to authorized users.

10. Workflow Engine and Orchestration Studio

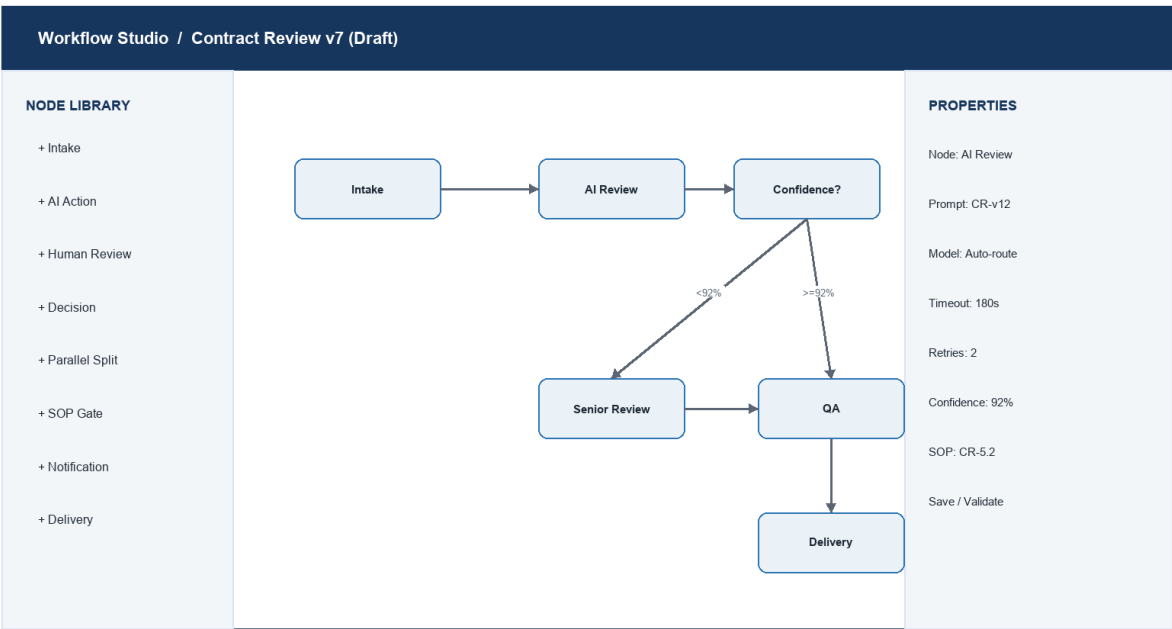


Figure 6. Legal Workflow Orchestration Studio wireframe

10.1 Node Types

Category	Examples
Triggers	Matter approved, document uploaded, timer, webhook, manual start
Actions	Create Job, assign user, transform document, call integration, generate deliverable
AI	OCR, classify, extract, draft, translate, research, verify, confidence gate
Human	Legal review, QA, compliance, approval, clarification, client approval
Control	Decision, parallel split/join, loop with limit, wait, SLA timer, escalation, subflow
SOP	Read instruction, checklist gate, evidence upload, attestation, training prerequisite
Events	Notification, Mattermost post, email, portal event, audit marker
Commercial	Quote gate, wallet reservation, consume/release, invoice handoff

10.2 Definition and Publication

ID	Requirement	Priority	Verification / acceptance
WFL-001	Administrators SHALL create workflows on a drag-and-drop canvas with searchable reusable nodes and typed connections.	P1	A valid sample workflow can be authored without code.

ID	Requirement	Priority	Verification / acceptance
WFL-002	The builder SHALL validate unreachable nodes, cycles without limits, missing branches, incompatible data types, absent error paths and unauthorized secrets.	P0	Invalid graphs cannot be published and show actionable errors.
WFL-003	Workflow definitions SHALL have Draft, Testing, Approved, Published, Retired states and immutable published versions.	P0	Editing Published creates a new Draft version.
WFL-004	Every execution SHALL pin the workflow, subflow, SOP, prompt, model-policy and rate versions used.	P0	Historical run remains reproducible after later edits.
WFL-005	The engine SHALL support conditional branches, bounded loops, parallel branches and deterministic joins.	P0	Automated scenarios prove branch and join behavior.
WFL-006	Each node SHALL define input/output schema, timeout, retry policy, idempotency, compensation/error path, owner and visibility.	P0	Runtime rejects schema-invalid payload and handles configured error path.
WFL-007	Long-running executions SHALL persist state and safely resume after worker restart.	P0	Chaos test restarts workers without duplicate side effects.
WFL-008	Timers and escalations SHALL survive deployment and use the Matter SLA calendar.	P0	Persisted timer fires once at the correct business time.
WFL-009	Client-specific overrides SHALL be constrained, validated, versioned and diffable against the base workflow.	P1	Override shows changed nodes and cannot bypass mandatory global controls.
WFL-010	Simulation SHALL execute with synthetic/redacted data and no production side effects while estimating time, cost and route.	P1	Simulation labels output and blocks external mutations.
WFL-011	Publish SHALL require peer approval for high-risk workflows and record change rationale and test evidence.	P0	Unauthorized self-publish is rejected.
WFL-012	Rollback SHALL select a prior published version for new executions; active executions remain pinned unless explicitly migrated.	P0	Rollback does not silently change in-flight cases.
WFL-013	Operational analytics SHALL report throughput, latency, cost, rework, error rate, SLA and bottleneck per node/version.	P1	Dashboard reconciles to execution records.
WFL-014	The studio SHOULD offer AI-assisted workflow generation, but generated graphs SHALL remain Draft until human validation and approval.	P2	AI cannot directly publish.

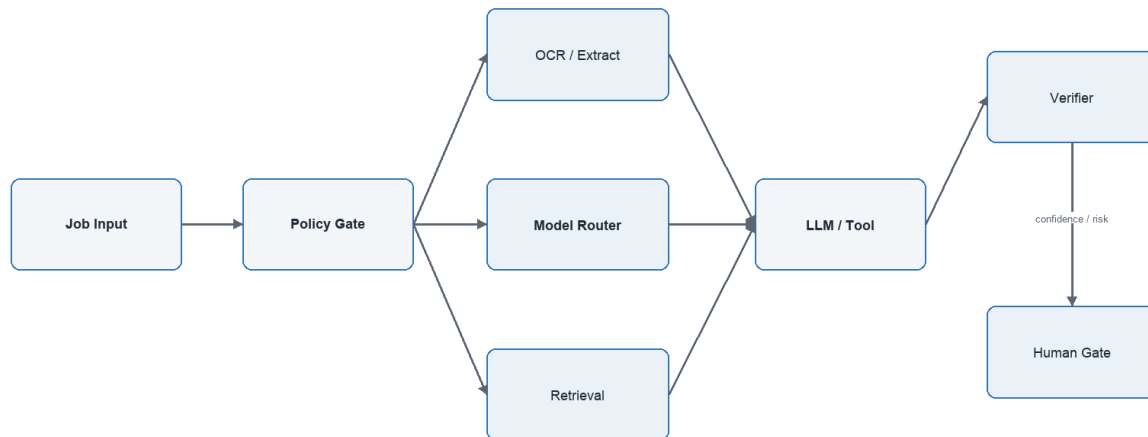
11. SOP Engine

11.1 SOP Model

An SOP is executable guidance, not a static attachment. It contains versioned instructions, prerequisites, role scope, ordered or conditional steps, checklist items, evidence requirements, expected duration, training links, exceptions, approvals and effective dates.

ID	Requirement	Priority	Verification / acceptance
SOP-001	SOP definitions SHALL be versioned with Draft, Review, Approved, Effective, Superseded and Retired states.	P0	Only an approved effective version may start new runs.
SOP-002	A workflow node SHALL bind to an exact SOP version or a controlled "latest approved" resolution stored at execution start.	P0	Execution record identifies immutable SOP version.
SOP-003	Mandatory steps SHALL block completion until required confirmation and evidence are present.	P0	UI and API reject bypass.
SOP-004	SOP steps SHALL support applicability conditions based on work type, jurisdiction, client, risk and document metadata.	P1	Conditional checklist produces expected resolved steps.
SOP-005	The engine SHALL capture performer, timestamps, evidence, comments, exceptions and reviewer decisions for every step.	P0	Complete evidence package can be exported.
SOP-006	Exceptions SHALL require reason, risk rating and approval according to policy.	P0	Unapproved exception cannot release a mandatory gate.
SOP-007	Users SHALL be notified when an SOP used by active work is superseded; migration policy SHALL be explicit.	P1	No silent mid-run procedure change.
SOP-008	SOP compliance analytics SHALL show completion, skipped/failed items, rework, error correlation and training gaps.	P1	Metrics trace back to SOP run records.
SOP-009	SOP changes SHALL support comparison and impact analysis across workflows, jobs and training.	P1	Dependency report lists affected published assets.

12. AI Orchestration and Governance



Every execution pins model, prompt, retrieval corpus, policy, cost, inputs, outputs and reviewer decisions.

Figure 7. Governed AI orchestration pipeline

12.1 AI Execution Record

Field group	Captured data
Identity	AI run ID, Client, Matter, Job, task, correlation ID
Configuration	Use case, model/provider/version, prompt version, policy version, tools
Inputs	Document versions/checksums, retrieval corpus snapshot, redaction and consent flags
Outputs	Raw and normalized result, citations, confidence, safety flags, checksum
Operations	Start/end, latency, tokens, provider cost, retries, fallback, error
Human feedback	Accepted/rejected/edited, reviewer, reasons, quality score, final comparison

12.2 Functional Requirements

ID	Requirement	Priority	Verification / acceptance
AI-001	All model calls SHALL pass through an authenticated AI gateway enforcing Client, purpose, data classification and provider policy.	P0	Direct provider access from application workers is blocked.
AI-002	The gateway SHALL route by use case, jurisdiction, confidentiality, capability, cost, latency and availability.	P0	Policy tests select/deny expected models.

ID	Requirement	Priority	Verification / acceptance
AI-003	Prompts SHALL be version-controlled, approved for production and referenced by immutable version in every run.	P0	Production call cannot use unapproved draft prompt.
AI-004	Sensitive input SHALL be minimized/redacted according to policy before leaving the approved boundary.	P0	DLP test detects and blocks prohibited payload.
AI-005	Provider data-retention and training settings SHALL match the Client contract and selected policy.	P0	Policy evidence is stored with provider configuration.
AI-006	AI outputs SHALL be treated as untrusted and validated for schema, citations, prompt injection, malware links and policy violations.	P0	Adversarial test cases are blocked or flagged.
AI-007	High-risk legal outputs SHALL require human legal review; confidence alone SHALL NOT bypass mandatory policy gates.	P0	High-confidence test still routes to required reviewer.
AI-008	Confidence SHALL include method and calibration version and SHALL not be displayed as factual certainty.	P1	UI shows confidence provenance and limitation.
AI-009	Citation verification SHALL link claims to accessible sources and record verification outcome.	P0	Unsupported citations are flagged before release.
AI-010	Retries and provider fallbacks SHALL be bounded, idempotent and preserve a complete attempt history.	P0	Failure simulation creates one logical result and all attempts.
AI-011	The system SHALL record token usage, provider cost and allocated internal cost per Client, Matter, Job and workflow node.	P1	Cost totals reconcile to provider reports within tolerance.
AI-012	Client data SHALL not enter shared learning corpora or model training without explicit approved authorization.	P0	Knowledge ingestion requires consent/policy gate.
AI-013	Human corrections SHOULD feed governed evaluation datasets with provenance, access control and opt-out policy.	P1	Dataset item links source, consent and reviewer.
AI-014	Model and prompt releases SHALL pass evaluation thresholds for accuracy, hallucination, citation, privacy, bias, cost and latency.	P0	Release is blocked below configured threshold.
AI-015	The platform SHALL support a kill switch by provider, model, prompt, use case, Client and jurisdiction.	P0	Disabled route is rejected immediately and audited.

13. Dashboards and User Experiences

13.1 Shared Design System

Internal dashboards use a consistent shell: left navigation, global search, KPI strip, role-specific primary workspace, right-side alerts/assistant panel and contextual action bar. The client portal uses the same visual language but replaces ERP vocabulary with client-centered labels.

Workspace	Primary users	Core modules
1. Executive Command Center	CEO / leadership	Revenue, clients, pipeline, SLA, margin, risk, AI ROI, quality, cash and forecasts
2. Operations Control Tower	Operations Manager	Intake, production pipeline, capacity, assignments, deadlines, queues, escalations and delivery
3. Senior Legal Workbench	Senior Associates	Review queue, documents/redlines, AI verification, SOP gates, playbooks, research and approvals
4. Junior Legal Workstation	Junior Associates	Today's work, guided SOP wizard, AI draft, editor, checklist, quality meter and escalation
5. Quality Command Center	QA	Pending QA, legal risk, scorecards, calibration, errors, client/jurisdiction quality and rework cost
6. Client Legal Operations Portal	Client users	Matters, submit work, documents, approvals, messages, reports, billing, wallet and settings
7. Sales & Client Success	Sales / relationship	Leads, opportunities, quotes, onboarding, LexPack sales, health, renewals and upsell
8. Platform Administration	System administrators	Identity, workflow, AI, prompts, integrations, queues, security, audit, storage and continuity

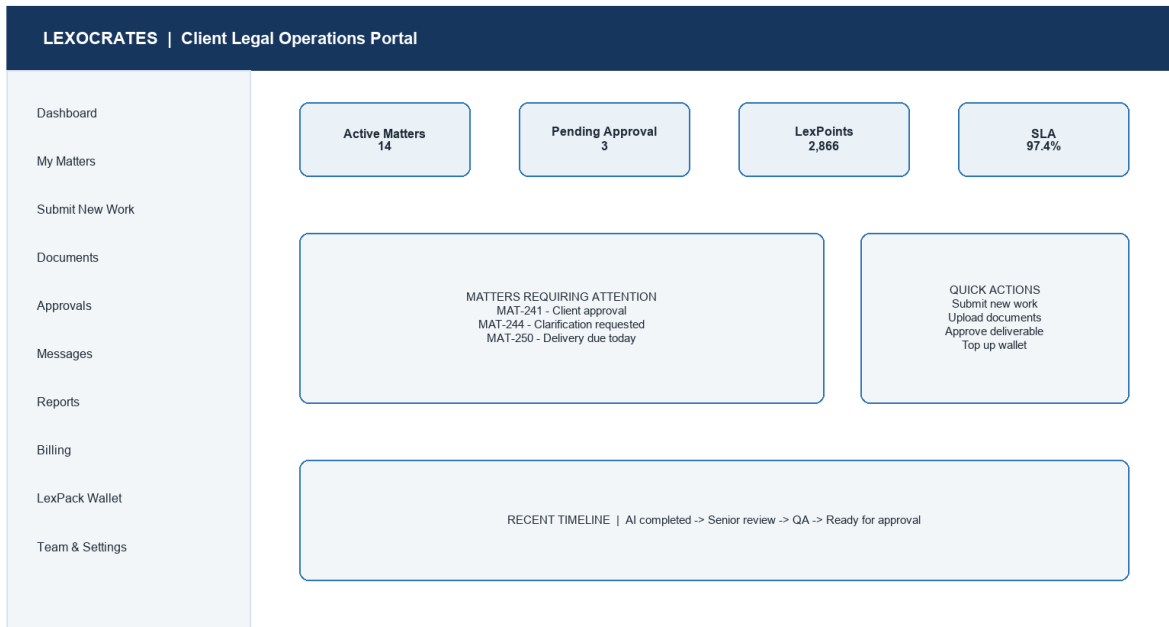


Figure 8. Client portal dashboard wireframe

13.2 Dashboard Requirements

ID	Requirement	Priority	Verification / acceptance
UI-001	Every dashboard SHALL display data permitted for the current role and Client/matter scope only.	P0	Role-based snapshot tests show no unauthorized records.
UI-002	KPIs SHALL state definition, time window, timezone, refresh time and drill-down source.	P1	Tooltip and drill-down reconcile to record list.
UI-003	The Junior Workstation SHALL present the next valid SOP step and minimize unrelated navigation.	P0	Usability test completes standard Job without module hunting.
UI-004	Senior and QA workspaces SHALL support side-by-side original, AI, human and final version comparison.	P1	Reviewer can trace and accept/reject changes.
UI-005	The Client Portal SHALL use "Client", "Matter", "Job/Work Item" only where meaningful and SHALL not expose ERPNext DocType labels.	P0	Client-facing terminology audit passes.
UI-006	Portal users SHALL be able to submit work, upload securely, track timeline, approve deliverables, view permitted billing/wallet data and message the team.	P0	End-to-end portal acceptance scenario passes.
UI-007	Client Finance Users SHALL see finance objects without confidential matter documents unless separately granted.	P0	Finance-only role test passes.
UI-008	Operations SHALL view platform health and production status but SHALL NOT administer secrets, identities or infrastructure.	P0	Restricted controls absent and API access denied.
UI-009	Dashboards SHALL provide accessible keyboard navigation, visible focus, text alternatives and non-color status cues.	P0	WCAG 2.2 AA automated/manual checks meet agreed threshold.
UI-010	Tables and feeds SHALL support filtering, saved views,	P1	UX test covers each state.

ID	Requirement	Priority	Verification / acceptance
	pagination/export controls and clear empty/error/loading states.		
UI-011	Critical actions SHALL require confirmation, show consequences and return a durable receipt/reference.	P0	Accidental double-submit does not duplicate action.
UI-012	All user-visible dates, currencies and numbers SHALL respect configured locale while storing canonical values.	P0	Locale matrix test passes.

14. Communications, Notifications and Collaboration

ID	Requirement	Priority	Verification / acceptance
COM-001	Each Matter SHALL support a linked collaboration space with stable identifiers stored in ERPNext.	P1	Matter link opens correct authorized Mattermost channel.
COM-002	Mattermost provisioning SHALL be idempotent and mirror authorized membership without granting broader tenant access.	P0	Retry creates no duplicate channel/team; revoked user loses access.
COM-003	Material decisions made in chat SHALL be capturable as ERP timeline/audit references.	P1	Selected message permalink and metadata are preserved.
COM-004	Notifications SHALL support portal, email and Mattermost channels with templates, preferences and mandatory security exceptions.	P0	Channel preference respected except mandatory alerts.
COM-005	Every notification SHALL have event ID, recipient, channel, template version, status and deduplication key.	P0	Retries do not produce duplicate notifications beyond policy.
COM-006	Sensitive content SHALL not be placed in email subject/body when policy requires authenticated portal access.	P0	Template security test finds links instead of confidential payload.
COM-007	Escalations SHALL target role/roster, not hard-coded individuals, and support acknowledgement and fallback.	P1	Unavailable primary triggers configured fallback.

15. API and Integration Requirements

15.1 API Standards

ID	Requirement	Priority	Verification / acceptance
API-001	APIs SHALL be versioned, documented with OpenAPI and use consistent error envelopes and correlation IDs.	P0	Contract validation and generated documentation pass.
API-002	Create/update APIs SHALL support idempotency keys where retries could duplicate financial, workflow or delivery effects.	P0	Repeated request returns same logical result.
API-003	List APIs SHALL enforce authorization before filtering/pagination and define stable sort order.	P0	No record-count or search leakage across tenants.
API-004	Webhooks SHALL be signed, timestamped, replay-protected, retryable and observable.	P0	Invalid/replayed signature is rejected.
API-005	Integration credentials SHALL be stored in a secrets manager and SHALL NOT appear in DocType fields, logs or exports.	P0	Secret scanning and access tests pass.
API-006	n8n SHALL perform external automation; core authorization, wallet posting, approval and workflow state SHALL remain in Lexocrates services.	P0	Architecture test proves n8n cannot directly bypass domain rules.
API-007	Connectors SHALL implement rate limits, timeout, circuit breaker, retry/backoff and dead-letter handling.	P0	Provider outage does not exhaust workers or lose requests.
API-008	Inbound files SHALL be quarantined until malware scanning and validation complete.	P0	Malicious test file cannot be opened or processed downstream.
API-009	Data exports SHALL be authorized, watermarked/logged where configured, encrypted in transit and time-limited.	P0	Expired export link fails and event is audited.

15.2 Integration Responsibilities

Integration	Primary use	Boundary rule
n8n	External orchestration/adapters	No direct wallet posting or permission bypass
Mattermost	Matter collaboration	Membership mirrors ERP authorization
AI providers	Model inference	Only through AI gateway
Object storage	Versioned encrypted files	Metadata/authorization remain in ERP
Payment gateway	LexPack/quote payments	Signed webhooks and exactly-once posting
Email/SMS	Notifications and verification	No prohibited confidential content
Legal research providers	Authorized research/citations	License and jurisdiction controls
Identity provider	SSO/MFA/SCIM	ERP maps identity to Client/roles

16. Audit, Security, Privacy and Compliance

16.1 Audit Event Schema

Field	Requirement
Identity	Event ID, event type, schema version, correlation/causation IDs
Actor	User/service, role, Client context, delegated/support-access context
Target	Object type/ID, Client, Matter, Job and field set where relevant
Action	Before/after or cryptographic reference, reason, approval, policy/version
Environment	Timestamp UTC, source IP/device/session, service/release
Integrity	Append-only storage, hash/signature or tamper-evident control, retention class

ID	Requirement	Priority	Verification / acceptance
AUD-001	The platform SHALL record authentication, authorization changes, reads/downloads of sensitive data, writes, approvals, financial postings, workflow/SOP/prompt changes, AI calls and deliveries.	P0	Audit coverage tests generate expected event types.
AUD-002	Audit events SHALL be append-only, time-synchronized and protected from alteration by ordinary administrators.	P0	Admin cannot edit/delete; integrity verification succeeds.
AUD-003	Audit search SHALL be access-controlled, exportable and capable of correlating a Matter from intake through delivery.	P0	Auditor reconstructs sample chain using correlation IDs.
AUD-004	Sensitive values and secrets SHALL be redacted from logs while preserving useful evidence.	P0	Log scanning finds no secret/test personal payload.
AUD-005	Retention SHALL be configurable by data class, jurisdiction, contract, legal hold and disposal policy.	P0	Legal hold prevents deletion; expiry creates disposal evidence.
AUD-006	The system SHALL support access, correction, export and deletion/anonymization workflows subject to legal exceptions.	P0	Privacy request test produces accountable decision and evidence.

16.2 Security Controls

- MFA for internal users and client administrators; risk-based MFA for all portal users.
- SSO via OIDC/SAML where configured; SCIM may provision enterprise users.
- TLS 1.2+ in transit and strong encryption at rest; keys separated by environment and rotated.
- Secure headers, CSRF protection, input validation, output encoding, dependency scanning and hardened file handling.
- Central secrets management; least-privilege service identities; no shared administrator accounts.
- Network segmentation, WAF/rate limiting, monitored privileged access and egress control.
- Backups encrypted, immutable where feasible, routinely restored and isolated from production credentials.

- Incident response runbooks for account compromise, data exposure, ransomware, AI-provider breach and integrity failure.

16.3 Compliance Position

The design supports controls commonly required for ISO/IEC 27001-aligned management systems and privacy obligations such as GDPR and PIPEDA. Compliance is not conferred by software alone. Lexocrates must define legal bases, contracts, records of processing, retention schedules, transfer mechanisms, incident procedures and evidence ownership with qualified counsel and auditors.

17. Reporting, Analytics and Observability

ID	Requirement	Priority	Verification / acceptance
OBS-001	Operational metrics SHALL cover intake, queue depth, cycle time, SLA, backlog, capacity, rework and delivery.	P0	Metrics reconcile to sampled records.
OBS-002	Quality metrics SHALL cover legal risk, error categories, SOP compliance, reviewer calibration, client/jurisdiction trends and cost of poor quality.	P1	Filters and denominator definitions are visible.
OBS-003	AI metrics SHALL cover latency, failures, fallback, token/cost, confidence calibration, citation accuracy, human acceptance and drift.	P0	Model/prompt/version filters work.
OBS-004	Commercial metrics SHALL separate quoted revenue, LexPack sales, wallet consumption, reservations, margin and outstanding receivables.	P0	Finance reconciliation meets agreed tolerance.
OBS-005	Platform telemetry SHALL include traces, metrics and structured logs sharing a correlation ID.	P0	A failed request is traceable across API, queue and integration.
OBS-006	Alerts SHALL be actionable, severity-ranked, deduplicated and linked to runbooks.	P0	Alert drill routes owner and demonstrates recovery action.
OBS-007	Client-visible reports SHALL exclude internal notes, costs, model details and other tenants unless explicitly approved.	P0	Data-leakage report tests pass.
OBS-008	Report definitions SHALL be versioned with metric definitions and effective dates.	P1	Historical report identifies definition version.

18. Non-Functional Requirements

ID	Requirement	Verification
NFR-PERF-001	Authenticated portal pages SHOULD reach p95 server response below 1.5 seconds for standard reads under agreed reference load.	Performance test
NFR-PERF-002	Standard transactional API requests SHOULD complete at p95 below 750 ms excluding external providers.	Load test
NFR-SCALE-001	The architecture SHALL horizontally scale stateless web and worker tiers and partition queues by workload class.	Scale test / architecture review
NFR-AVAIL-001	Production target availability SHALL be at least 99.9% monthly excluding approved maintenance; higher tiers may be contracted.	Monitoring report
NFR-DR-001	Baseline RPO SHALL be ≤ 15 minutes and RTO ≤ 4 hours, subject to approved business-impact analysis.	Restore/failover exercise
NFR-REL-001	Financial and externally mutating operations SHALL be exactly-once in effect through idempotency and reconciliation.	Fault-injection test
NFR-SEC-001	Critical/high security findings SHALL block release unless formally risk-accepted by authorized owners.	Release evidence
NFR-PRIV-001	Data minimization, purpose limitation, retention and tenant isolation SHALL be verified before production.	Privacy/security review
NFR-ACC-001	Client and core internal interfaces SHALL target WCAG 2.2 AA.	Automated/manual accessibility test
NFR-COMP-001	The UI SHALL support current Chrome, Edge, Safari and Firefox enterprise versions as agreed.	Browser matrix
NFR-MAINT-001	Domain, API, workflow, SOP and configuration changes SHALL include automated tests and migration/rollback notes.	CI gate
NFR-OBS-001	100% of synchronous requests and asynchronous jobs SHALL carry correlation identifiers.	Telemetry audit
NFR-I18N-001	Timestamps SHALL be stored in UTC and rendered in user timezone; monetary values SHALL include currency.	Locale/timezone test
NFR-DATA-001	Backups SHALL be encrypted, retention-controlled and restore-tested at least quarterly.	Restore report
NFR-USAB-001	A trained portal user SHOULD submit standard new work in under five minutes without staff help.	Moderated usability test
NFR-PORT-001	Business rules SHALL be isolated from UI and external automation to permit API and channel reuse.	Architecture review

18.1 Capacity Baseline

Before sizing, product owners must approve expected Clients, active users, concurrent sessions, matters/day, documents/day, average/max file size, OCR pages/day, AI calls/day, retention years and reporting concurrency. Load tests shall use at least 1.5× the approved peak and include failure/recovery behavior.

19. ERPNext / Frappe Implementation Guidance

19.1 Recommended DocTypes

DocType	Approach	Notes
Lex Client Profile	Custom / extend Customer	Tenant root; do not rely on Customer permissions alone
Lex Contact Role	Custom child/link	Contact function and notification preferences
Lex Portal Membership	Custom	User ↔ Client, role template, status, dates
Lex Matter	Custom	Core legal assignment and tenant scope
Lex Matter Access	Custom child	User/group matter grants
Lex Job	Custom	Executable unit, workflow/SLA snapshot
Lex Task	Custom	Human/system task instance
Lex Document Record	Custom	File metadata, version, checksum, classification
Lex Deliverable	Custom	Approved release package and receipt
Lex Wallet	Custom	Balance container; computed/reconciled
Lex Wallet Transaction	Custom, immutable	Posted ledger credit/debit
Lex Wallet Reservation	Custom	Available-balance hold with expiry/release policy
Lex Workflow Definition/Version	Custom	Graph metadata and immutable version payload
Lex Workflow Execution/Node Run	Custom	Durable runtime state
Lex SOP / SOP Version / SOP Run	Custom	Executable procedure and evidence
Lex Prompt / Prompt Version	Custom	Governed AI prompt lifecycle
Lex AI Execution	Custom	Model call provenance, cost and feedback
Lex Approval	Custom	Generic multi-stage approval evidence
Lex Audit Event	Custom/external WORM	Correlated tamper-evident events
Lex SLA Policy / Clock	Custom	Calendar-aware targets and timers

19.2 ERPNext Mapping

- Use ERPNext Customer, Address, Contact, Quotation, Sales Order, Sales Invoice, Payment Entry and accounting dimensions where their semantics fit.
- Do not overload ERPNext Project/Task if required legal states, tenant permissions, evidence and versioned workflow semantics cannot be enforced cleanly. Prefer purpose-built Lex Matter and Lex Job linked to financial documents.
- Use Frappe User for authentication but keep client membership/authorization in a dedicated DocType with server-side permission hooks and query conditions.

- Use naming series for display IDs while retaining immutable UUIDs for APIs and correlation.
- Mark ledger, audit and published version DocTypes as non-editable after posting/publication; implement reversal/new-version patterns.
- Use background jobs for OCR, AI, conversion, indexing, report generation and external connectors; keep request handlers short.
- Store large documents in encrypted object storage; Frappe File retains authorized metadata and secure references rather than public URLs.

19.3 Hooks and Services

Layer	Guidance
Validation	DocType validate/before_submit for local invariants; domain service for cross-record transactions
Authorization	has_permission, permission_query_conditions, service-layer checks and explicit tenant scope
Events	Publish transactional outbox after commit; consumers are idempotent
Financial	Use database transaction and row locking for reservation/posting; scheduled reconciliation
Workflow	Dedicated durable runtime; do not model complex graph execution solely with ERPNext Workflow states
Audit	Central event writer invoked from domain services, permission changes, downloads and integrations
API	Whitelisted methods call domain services; schema validation and stable versioning
Testing	Unit, DocType, API, permission, migration, workflow, financial concurrency and end-to-end suites

19.4 Migration and Customization Rules

- All schema/config changes ship as versioned patches with forward verification and documented rollback/compensation.
- Fixtures are limited to controlled reference configuration; production secrets and tenant records are never fixtures.
- Avoid client scripts for authoritative validation. Mirror helpful client-side checks with server enforcement.
- Custom apps use clear module boundaries: core domain, portal/API, workflow/SOP, AI, integrations and reporting.
- Release notes identify DocType, workflow, permission, data migration and operational impacts.

20. Data Model Dictionary

Entity	Key fields	Critical constraints
Client	client_id (UUID), display_id, legal_name, status, country, currency, timezone, risk_class, created_at	Unique legal name/country review; immutable ID
PortalMembership	membership_id, client_id, user_id, roles, status, valid_from/to, department	Unique active membership per user/client
Matter	matter_id, client_id, title, status, billing_method, jurisdiction, SLA, classification	Client immutable; billing method enum
Job	job_id, matter_id, client_id, work_type, status, assignee, workflow_version, due_at	Client must equal Matter client
DocumentVersion	document_id, version_id, client_id, matter_id, storage_key, checksum, classification	Checksum and storage key unique per version
Wallet	wallet_id, client_id, unit, status	One active per client/unit
WalletTransaction	txn_id, wallet_id, matter_id, job_id, type, amount, posted_at, idempotency_key	Non-zero signed amount; immutable posted row
WalletReservation	reservation_id, wallet_id, matter_id/job_id, amount, status, expires_at	Active reservations included in available balance
WorkflowVersion	workflow_id, version, graph_hash, status, approved_by, published_at	Published graph immutable
WorkflowExecution	execution_id, workflow_version, subject_type/id, status, started/ended	Pinned versions and correlation
SOPRun	sop_run_id, sop_version, job_id, status, evidence_manifest	Cannot complete with unmet mandatory items
AIExecution	ai_run_id, client/matter/job, model, prompt_version, input/output_hash, cost, decision	Policy and provenance required
Deliverable	deliverable_id, job/matter, version_set, approval_state, delivered_at	Release gates satisfied
AuditEvent	event_id, client, actor, action, target, occurred_at, correlation_id, integrity	Append-only; UTC timestamp

20.1 Data Integrity Rules

- Client ID is immutable after creation and duplicated onto high-volume child records to simplify secure filtering.
- Cross-Client links are rejected unless explicitly designated global reference data.
- All financial quantities use fixed-precision decimal types; never binary floating point.
- All timestamps are timezone-aware UTC; user timezone is a presentation concern.
- Enumerations are controlled reference values with migration strategy, not free text.
- Soft deletion is not a substitute for retention/disposal; deletion state and evidence are explicit.
- Search indexes and analytics stores are downstream projections and must be re-creatable from authoritative records.

21. End-to-End Sequence and Acceptance Scenarios

21.1 Scenario A - New Client, Quoted Matter

Step	Actor / system	Expected result
1	Applicant	Submits organization and primary-contact registration.
2	Platform	Creates pending Client and review case; no Matter or wallet credit.
3	Compliance / Sales	Completes conflict/KYC/commercial checks and approves.
4	Identity	Primary Client Administrator accepts invitation and enrolls MFA.
5	Client user	Submits contract-review work and files.
6	Operations / Finance	Creates quote; client accepts exact version.
7	Platform	Creates/open Matter, Jobs and pinned workflow/SOP versions.
8	AI + legal team	Processes, reviews, verifies citations and completes QA/compliance gates.
9	Client approver	Approves deliverable or requests versioned rework.
10	Platform / Finance	Delivers exact file versions, invoices and closes when conditions pass.

21.2 Scenario B - Existing Client, Wallet Matter

Step	Expected result
1	Client purchases LexPack; signed payment event posts one wallet credit.
2	Client submits new Matter with Wallet billing.
3	Platform reserves estimated LexPoints atomically; available balance decreases.
4	Jobs execute; each approved consumption posts a debit with rate version and references.
5	Unused reservation is released; statement reconciles posted and available balances.
6	Balance remains indefinitely; alerts recommend top-up without expiry language.

21.3 Scenario C - Workflow Failure and Recovery

Step	Expected result
1	AI provider times out; bounded retry records attempt.
2	Circuit breaker routes to permitted fallback or human triage.
3	Worker restarts; execution resumes from persisted node without duplicate debit or delivery.

Step	Expected result
4	Operations sees alert, correlation trace and runbook link.
5	Resolution and any override are audited; SLA timer follows policy.

22. Verification Strategy and Traceability

22.1 Test Levels

Level	Coverage
Unit	Domain rules, calculations, transitions, policies and serializers
Integration	ERPNext DocTypes, database transactions, queues, object storage, identity, AI gateway
Contract	OpenAPI, webhook signatures, provider adapters and event schemas
Permission	Role × tenant × matter × classification matrix and negative tests
End-to-end	Registration through delivery for quoted and wallet scenarios
Security	SAST/DAST, dependency, secrets, tenant isolation, file attacks, abuse and penetration testing
Performance / resilience	Reference load, queue saturation, provider failure, restart, failover and restore
UX / accessibility	Role workflows, keyboard/screen reader, localization and error recovery
Financial	Concurrency, idempotency, reconciliation, reversal and rounding
AI evaluation	Golden sets, adversarial inputs, citations, privacy, calibration, drift and human acceptance

22.2 Requirement Traceability Matrix

Requirement family	Design component	Primary verification	Owner
REG / USR	Onboarding + Identity	E2E, permission, security	Product / IAM
WAL	Commercial + ledger	Unit, concurrency, reconciliation	Finance / Engineering
MAT / JOB	Matter production	E2E, state, SLA	Operations / Engineering
WFL / SOP	Orchestration	Graph validation, recovery, evidence	Platform / QA
AI	AI gateway	Evaluation, security, cost reconciliation	AI / Legal / Security
UI	Portal and workspaces	UX, accessibility, permission	Product / QA
COM / API	Integrations	Contract, retry, security	Integration Engineering
AUD / SEC	Governance	Audit, penetration, privacy	Security / Compliance

Requirement family	Design component	Primary verification	Owner
OBS / NFR	Platform quality	Load, resilience, DR, observability	SRE / QA

22.3 Definition of Done

- Requirement and acceptance criteria implemented and linked to tests.
- Permission and tenant-isolation negative tests included.
- Audit events and operational telemetry verified.
- Data migration and rollback/compensation documented.
- Security/privacy review completed for affected data flows.
- User-facing text, accessibility and error states reviewed.
- Runbook, dashboards and alerting updated for operational changes.
- Product owner and relevant domain owner accept evidence.

23. Delivery Roadmap

Phase	Scope	Exit criteria
0. Foundations	Domain vocabulary, tenant model, identity, CI/CD, environments, audit/event patterns	Threat model and architecture approved; permission harness passing
1. Client & Intake	Registration, onboarding, Contacts, Portal Users, client portal shell	Active Client and multi-user lifecycle passes E2E
2. Commercial & Matters	Quotes, wallet ledger/reservations, Matter/Job/document foundations	Quoted and Wallet scenarios reconcile and are secure
3. Production Runtime	Workflow engine, SOP engine, assignments, SLA, workstations	Pinned versioned execution survives faults
4. AI & Quality	AI gateway, prompt governance, evaluation, Senior/QA workspaces	Governed AI release thresholds and human gates pass
5. Integrations & Reporting	Mattermost, n8n, payment, research, dashboards/reports	Connector resilience and reporting reconciliation pass
6. Enterprise Hardening	SSO/SCIM, DR, observability, retention, privacy workflows, scale	Security review, restore drill, load and operational readiness sign-off

23.1 Release Gates

- No open critical/high security defect without documented authorized acceptance.
- Zero known cross-tenant data exposure; permission regression suite passes.
- Wallet reconciliation and idempotency tests pass with no unexplained variance.
- Backup restore and worker-recovery tests pass in production-like environment.
- AI use cases meet approved evaluation thresholds and mandatory human gates.
- Operations, security, finance and support runbooks are exercised.

24. Risks, Assumptions and Open Decisions

ID	Type	Item	Required decision / mitigation
RA-01	Assumption	ERPNext/Frappe remains the transactional core.	Revalidate after scale and workflow-runtime spike.
RA-02	Risk	Complex graph runtime implemented only with ERPNext Workflow.	Use dedicated durable execution model.
RA-03	Risk	Client data leaks through search, exports, files or background jobs.	Server-side scope everywhere; adversarial tenant tests.
RA-04	Risk	Wallet race conditions or duplicate payment events.	Row locking, idempotency, immutable ledger, reconciliation.
RA-05	Risk	AI hallucination or unsupported citations.	Evaluation, verifier, citations and mandatory human gates.
RA-06	Open	Exact LexPoint consumption and reservation policy.	Finance/product approve rate, rounding and cancellation rules.
RA-07	Open	Data residency and provider allowlists per market.	Privacy/security/legal decision before client contract.
RA-08	Open	RPO/RTO and availability by service tier.	Business-impact analysis and customer commitments.
RA-09	Open	Trust balance/retainer accounting scope.	Exclude until finance/legal requirements are approved.
RA-10	Open	Eight dashboard KPI definitions and thresholds.	Metric catalog with owners and denominator rules.
RA-11	Risk	Mattermost membership drift from ERP authorization.	Automated reconciliation and rapid revocation.
RA-12	Risk	Workflow or SOP changes affect active work unexpectedly.	Immutable snapshots and explicit migration workflow.

Appendix A. Status and Transition Catalog

Object	Core states	Notes
Client	Draft, Submitted, Under Review, Needs Info, Approved, Provisioning, Active, Suspended, Closed, Rejected	All transitions authorized and audited
Portal User	Invited, Active, Suspended, Locked, Revoked	Contact remains distinct
Matter	Draft, Intake Review, Awaiting Quote, Awaiting Funding, Open, On Hold, Delivery Pending, Completed, Closed, Cancelled	Billing gates before Open
Job	Draft, Ready, Queued, Running, Human Review, QA, Compliance, Ready to Deliver, Delivered, Closed, Failed, Cancelled	Return/retry paths preserve history
Workflow Version	Draft, Testing, Approved, Published, Retired	Published immutable
Workflow Execution	Pending, Running, Waiting, Paused, Completed, Failed, Cancelled	Pinned versions
SOP Version	Draft, Review, Approved, Effective, Superseded, Retired	Effective-date aware
Deliverable	Draft, Review, QA, Compliance, Client Approval, Approved, Delivered, Rejected, Superseded	Exact file-version set
Wallet Transaction	Draft, Approved, Posted, Reversed	Posted immutable
AI Run	Queued, Running, Verifying, Human Review, Completed, Failed, Cancelled	All attempts retained

Appendix B. Event and Naming Catalog

Pattern	Example	Rule
Client	CLI-2026-000145	Human-readable series; UUID is API identity
Matter	MAT-2026-000089	Unique and never reused
Job	JOB-2026-000341	Unique and linked to Matter
Wallet transaction	WTX-2026-001245	Immutable posting ID
Workflow execution	WEX-<UUID>	Globally unique correlation target
Audit event	AEV-<UUID>	Append-only event identity
Event name	matter.created.v1	Past-tense domain event with schema version
Idempotency key	client/system supplied UUID	Scoped to operation and retained beyond retry window

Appendix C. Sample API Contracts

Representative request - create Matter (fields abbreviated):

```
{
  "client_id": "<uuid>",
  "title": "Review 25 NDAs",
  "billing_method": "WALLET",
  "jurisdiction": ["CA-ON"],
  "priority": "HIGH",
  "requested_due_at": "2026-08-15T12:00:00Z",
  "documents": [{"version_id": "<uuid>"}]
}
```

Representative error envelope:

```
{
  "error": {"code": "INSUFFICIENT_WALLET_BALANCE", "message": "Top up or
choose quoted-price billing.", "details": {"required": "120.00", "available":
"82.00"}},
  "correlation_id": "<uuid>"
}
```

- Do not expose internal stack traces, SQL, secrets or other tenant identifiers.
- Return stable machine codes and localized human messages.
- Use decimal strings for financial/LexPoint quantities.
- Require Idempotency-Key for payment, wallet, workflow-start and delivery mutations.

Appendix D. Security and Go-Live Checklist

#	Go-live control	Owner	Evidence / sign-off
1	Tenant-isolation threat model and negative permission suite approved		
2	MFA, SSO, session expiry, lockout and recovery tested		
3	Secrets migrated to approved manager and rotated		
4	TLS, headers, WAF/rate limits and network segmentation verified		
5	File quarantine, malware scanning, content validation and download authorization tested		
6	Encryption at rest and key-rotation procedures verified		
7	Wallet concurrency, idempotency, reconciliation and reversal tested		
8	AI provider policy, redaction, retention, training opt-out and kill switches verified		
9	Audit completeness, integrity, clock sync and privileged-read logging verified		
10	Retention, legal hold, data export and disposal workflows tested		
11	Backup restore, RPO/RTO, failover and incident exercises completed		
12	Accessibility, localization and browser matrix passed		
13	Operational dashboards, alerts, runbooks and on-call ownership active		
14	Release/migration rollback evidence approved		
15	Client-facing terms and privacy/legal notices approved		

Appendix E. Glossary

Term	Definition
AI Gateway	Controlled service through which all model requests pass.
Client Administrator	Portal user authorized to manage the Client's own users and settings.
Conflict Check	Review to identify conflicts of interest before accepting work.
DLP	Data loss prevention controls that detect/restrict sensitive data movement.
Idempotency	Property ensuring retries cause one logical effect.
Legal Hold	Suspension of normal disposal for evidence or litigation need.
LexPoint	Non-expiring unit of service capacity held in a Wallet; not an hour.
Matter-level Permission	Access grant restricting a user to specified Matters/actions.
RPO / RTO	Maximum acceptable data loss / restoration time objectives.
SLA	Service-level target and associated calendar/escalation rules.
Tenant	Logical data/security boundary of one Client organization.
WORM	Write once, read many; a tamper-resistant storage approach.

Appendix F. Maintenance Instructions

- Assign a document owner and review at least quarterly or before each major release.
- Record accepted change requests in Document Control and update impacted requirement IDs, diagrams, DocTypes, tests and risks.
- Never renumber baselined requirements. Add new IDs at the end of the relevant family.
- Mark superseded requirements Deprecated with replacement references; do not delete history.
- Export a dated PDF at each approved baseline and store DOCX/PDF with the corresponding source-control release tag.
- Review external standards, laws, provider capabilities and ERPNext versions separately; this SRS deliberately avoids claiming automatic compliance.

END OF BASELINE Version 1.0 is ready for stakeholder review. Open decisions in Section 24 must be assigned owners and resolved before affected functionality is released to production.